

Annular H^∞ and Hardy Criteria for the Direct Weighted Prefix

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Abstract

The remaining determinant bridge is an absolute weighted coefficient norm, but it need not be proved coefficient by coefficient or root by root. We give two direct analytic criteria. If the logarithmic mismatch is small in H^∞ on any circle strictly outside the target disk, Cauchy's estimate controls the full weighted coefficient norm. A vanishing H^2 boundary norm on the same annulus also suffices by Cauchy–Schwarz. Both bounds are explicit. At radius 1.41, which lies between the target radius 1.4 and the certified deterministic radius 1.426787..., their constants are 139.007... and 8.292.... An endpoint polynomial family shows that vanishing H^2 norm on the target circle alone is insufficient. No actual noisy annular convergence is asserted.

1 Logarithmic mismatch

Let

$$g_\sigma(z) = \sum_{n \geq 2} \frac{\tau_{\sigma,n} - a_n}{n} z^n$$

be holomorphic on a neighborhood of the closed disk $|z| \leq \rho$, where $\rho > R > 0$. Define the full absolute coefficient norm

$$P_\sigma^\infty(R) = \sum_{n \geq 2} \frac{|\tau_{\sigma,n} - a_n| R^n}{n}.$$

2 Two annular criteria

Theorem 2.1. *Put $x = R/\rho < 1$.*

1. *If $\sup_{|z| \leq \rho} |g_\sigma(z)| \leq M_\sigma$, then*

$$P_\sigma^\infty(R) \leq M_\sigma \frac{x^2}{1-x}.$$

2. *If*

$$\|g_\sigma\|_{H^2(\rho)}^2 := \sum_{n \geq 2} \frac{|\tau_{\sigma,n} - a_n|^2 \rho^{2n}}{n^2} \leq H_\sigma^2,$$

then

$$P_\sigma^\infty(R) \leq H_\sigma \frac{x^2}{\sqrt{1-x^2}}.$$

In particular, $M_\sigma \rightarrow 0$ or $H_\sigma \rightarrow 0$ makes the full norm vanish. Every moving prefix is bounded by $P_\sigma^\infty(R)$, so the RH-292 shortened prefix follows.

Proof. Cauchy's coefficient estimate gives

$$\frac{|\tau_{\sigma,n} - a_n|}{n} \leq M_\sigma \rho^{-n},$$

and summing the geometric series proves the first bound. For the second, apply Cauchy–Schwarz to

$$\sum_{n \geq 2} \left(\frac{|\tau_{\sigma,n} - a_n| \rho^n}{n} \right) x^n.$$

The square sum of x^n from $n = 2$ onward is $x^4/(1 - x^2)$. □

3 Endpoint obstruction

Theorem 3.1. *The strict radius gap in the H^2 criterion cannot be removed uniformly. For every $N \geq 1$, let*

$$g_N(z) = \frac{1}{N} \sum_{n=2}^{N+1} (z/R)^n.$$

Then

$$\|g_N\|_{H^2(R)} = N^{-1/2} \longrightarrow 0, \quad \sum_{n=2}^{N+1} |[z^n]g_N| R^n = 1.$$

Proof. There are N coefficients, each with boundary-scaled modulus $1/N$. Their square sum is $1/N$, while their absolute sum is one. □

4 Certified annulus and protocol

The deterministic coefficient radius is $\rho_* = 0.85\lambda = 1.426787483864073\dots$. Taking $\rho = 1.41$ and $R = 1.4$ gives

$$\frac{x^2}{1-x} = 139.0070921986\dots, \quad \frac{x^2}{\sqrt{1-x^2}} = 8.2924678943\dots$$

The computation records these constants and the endpoint family. It does not estimate g_σ for the noisy operator.

Remark 4.1. The theorem is a reopening criterion, not a verified annular bridge. If activated it supplies the direct P_σ^∞ bound, but no such H^∞ or H^2 convergence is currently proved. Gates A–E remain false/open and no Hilbert–Polya, zeta-divisor, or RH statement follows.